

Arduino & Breadboard Circuit Guide: LEDs, Motors, Resistors & Diodes

Document: EDU-EE-ARDUINO-01 | Practical Electronics Lab Wiring Standard

1. Light Emitting Diodes (LEDs) and Current Limiting Resistors

Light Emitting Diodes (LEDs) are polarized semiconductor diodes. The longer lead is the Anode (positive terminal), and the shorter lead next to the flat notch on the plastic lens casing is the Cathode (negative terminal / GND). Under forward bias, current must be strictly limited to prevent thermal runaway and permanent destruction.

Forward Voltage (Vf) by Color: Red, Orange, Yellow Vf = 1.8V to 2.2V; Green Vf = 2.2V to 3.0V; Blue, Bright White Vf = 3.0V to 3.4V. Standard 5mm through-hole LEDs have a continuous forward current limit (If_max) of 20mA. For optimal brightness and component lifespan, design for 10mA to 15mA.

Resistor Formula: $R = (V_{supply} - V_f) / I_f$. For a 5V Arduino GPIO pin driving a standard Red LED (Vf=2.0V) at 15mA: $R = (5.0V - 2.0V) / 0.015A = 200 \text{ Ohms}$. The standard readily available resistor values are 220 Ohms (providing 13.6mA) or 330 Ohms (providing 9.1mA). NEVER connect an LED directly between an Arduino pin and GND without a series current-limiting resistor; doing so exceeds the 40mA absolute maximum pin limit of the ATmega328P and burns out the microcontroller output driver.

2. DC Motors, Servos, and Inductive Back-EMF Protection

Direct Microcontroller Connection Warning: Microcontrollers cannot drive DC motors directly. An Arduino or ESP32 GPIO pin can supply a maximum of 20mA to 40mA. Even small hobby DC motors draw 200mA to 500mA running and up to 2A at stall. Attempting to power a motor directly from a digital I/O pin causes immediate chip destruction or permanent port death.

Transistor and MOSFET Drivers: To control a DC motor from an Arduino, use an N-channel MOSFET (such as IRLZ44N, RFP30N06LE logic-level FETs) or an NPN power transistor (TIP120 Darlington or 2N2222 for micro-motors). Connect the Arduino PWM pin (e.g. Pin 9) through a 1k Ohm resistor to the Gate/Base. The Source/Emitter connects to GND, and the Drain/Collector switches the motor negative lead to GND.

Mandatory Flyback Protection Diode: Motors are highly inductive loads. When current is abruptly switched off by the transistor or PWM signal, the collapsing magnetic field creates a reverse voltage spike (inductive kickback / back-EMF) that can exceed 100V. A flyback rectifier diode (1N4001, 1N4007, or Schottky 1N5819) MUST be placed in parallel across the motor terminals with reverse polarity: Cathode (striped end) to the positive motor supply rail (+V), and Anode to the switched ground terminal. The diode clamps the high-voltage spike harmlessly across the coil.

3. Servo Motors (SG90, MG996R) Wiring & Power Distribution

Micro-servos like the SG90 require three connections: Brown/Black wire to Ground (GND), Red wire to 5V Power (+5V), and Orange/Yellow wire to a Microcontroller PWM pin (e.g., Pin 9 or 10 on Arduino). While a single SG90 can be tested from the Arduino 5V header, multi-servo assemblies or loaded servos must use an external 5V 2A power supply to avoid triggering Arduino brownout resets.

4. Breadboard Architecture, Power Rails, & Common Ground Rules

Common Ground Reference Rule: When using an external battery (e.g., 9V or 12V) or external power supply for motors, relays, or LEDs, the negative (-) terminal of the external supply MUST be physically connected to the Arduino GND pin. Without this common reference ground, logic voltage levels float unpredictably, causing false triggers, erratic sensor readings, and communication failure.

Breadboard Power Rails: Full-size 830-tie-point solderless breadboards feature continuous red (+) and blue/black (-) power distribution buses along the top and bottom edges. Important: Many commercial breadboards have split power rails in the center (look for a break in the colored line). Always bridge the middle split with jumper wires to ensure power reaches both halves of the breadboard.

5. Pushbutton Pull-Up and Pull-Down Resistor Configurations

Floating Input Prevention: A digital input pin connected to an open switch is in a high-impedance floating state, picking up electromagnetic noise and flipping randomly between 0 and 1. To fix this, use a 10k Ohm pull-up resistor connected between the pin and 5V (with switch to GND), or configure the internal hardware pull-up resistor in software via `pinMode(pin, INPUT_PULLUP)`.